

# CrowdGuard

Real-Time Crowd Safety Monitoring System

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**Team Innovators**

Dussehra Jumbo Savari | Safety Innovation Challenge 2026

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## Dussehra Jumbo Savari — A Beautiful Celebration, A Real Danger

**500K+**

**Spectators**

packed into narrow streets annually

**2,000+**

**Crowd Deaths**

in India over the past 20 years

**0**

**Real-Time Alerts**

at most religious events today

- Manual headcounts are slow, inaccurate, and strain police resources
- No automated system alerts when a zone exceeds safe crowd capacity
- Crowd density at Dussehra Jumbo Savari rises rapidly — reaction time is critical
- 2016 Varanasi Dussehra stampede: 24 lives lost due to bridge overcrowding
- Existing tech solutions require expensive hardware or cloud connectivity

INTRODUCING

# CrowdGuard

A real-time, fully offline crowd monitoring system that detects overcrowding using a standard camera and instantly alerts authorities — no internet, no expensive hardware required.

**Works on Any Laptop or IP Camera**

## Simple 4-Step Detection Pipeline

**01**

### Capture

Laptop or IP camera captures live video feed of the crowd area in real time



**02**

### Detect

OpenCV HOG algorithm scans every frame, drawing boxes around each detected person



**03**

### Count

Temporal smoothing stabilises the count across frames, eliminating false readings



**04**

### Alert

When count reaches threshold: red screen overlay + continuous audio alarm fires instantly

## Built for Real-World Outdoor Conditions



### Any Camera

Works with any laptop webcam or IP camera. No special hardware required.



### Real-Time Detection

Processes video at 15-30 FPS. Alerts within 1 second of threshold crossing.



### Dual Alert System

Visual red-border overlay AND continuous audio alarm for immediate attention.



### 100% Offline

No internet connection needed. Works in remote venues with no network.



### Configurable Threshold

Set crowd limit per zone. Change without touching code - just edit config.



### Outdoor Optimised

CLAHE preprocessing handles glare, shadows, and harsh outdoor lighting.

# CrowdGuard vs Manual Monitoring

| Capability        | Manual Monitoring            | CrowdGuard                       |
|-------------------|------------------------------|----------------------------------|
| Detection Speed   | x Minutes (manual count)     | ✓ < 1 second                     |
| Accuracy          | x Prone to human error       | ✓ NMS + smoothing = stable count |
| Cost              | x Many staff required        | ✓ One laptop + free software     |
| Alert Mechanism   | x Verbal or radio only       | ✓ Auto visual + audio alarm      |
| Internet Required | x No                         | ✓ No — fully offline             |
| Scalability       | x More staff per zone needed | ✓ Add camera per zone            |

## Social, Economic & Scale Impact

### SOCIAL IMPACT

- Prevent stampedes and crowd crush fatalities at religious events
- Faster emergency response through real-time overcrowding alerts
- Safer experience for lakhs of Dussehra Jumbo Savari devotees
- Builds public trust in event safety management systems

### ECONOMIC IMPACT

- Zero recurring cost — fully open-source software stack
- Replaces expensive proprietary CCTV analytics platforms
- Reduces legal liability and compensation costs for organizers
- Saves crores in rescue, recovery, and medical response costs

### SCALABILITY

- Deploy at temples, religious fairs, stadiums, and malls
- Multi-camera zone support planned for Phase 2 rollout
- Applicable across all major Indian festivals and public events
- Exportable model for global religious crowd management

# Who Buys CrowdGuard and How

## TARGET CUSTOMERS

### Municipal Corporations

Deploy at major public events and processions

### Religious Trusts & Temples

Kumbh Mela, Dussehra, Ganesh Utsav, Rath Yatra

### Event Management Firms

Concerts, trade fairs, and sports events

### Malls & Stadiums

Permanent crowd monitoring infrastructure

## REVENUE STREAMS

### Per-Event License

₹5,000 – ₹20,000/event

One-time deployment fee per event

### Annual SaaS Subscription

₹50K – ₹2L/year

Unlimited events, cloud dashboard, updates

### Hardware + Software Bundle

₹1.5L – ₹5L

Preconfigured device with 3-year support

### Government Tender

Project-based

Smart City and Kumbh Mela integration bids



# From Prototype to City-Wide Deployment

Now – 3 Months

3 – 9 Months

9 – 18 Months

## Phase 1

- Single-camera deployment
- HOG detection + NMS alerts
- Configurable zone threshold
- Pilot at local Dussehra event

## Phase 2

- Multi-camera zone monitoring
- Mobile app for police officers
- SMS / WhatsApp alert integration
- Optional cloud dashboard

## Phase 3

- AI model upgrade (YOLO option)
- Police control room integration
- Predictive crowd flow analysis
- State government rollout

OUR TEAM

# Team Innovators

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- Passionate about using technology to solve real public safety challenges
- Built CrowdGuard end-to-end — from concept to a working prototype
- Expertise in computer vision, Python, and real-world event deployment
- Driven by the mission of zero preventable crowd fatalities in India

# Thank You

Let's Make Every Crowd Event Safer

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**Problem** | Dussehra Jumbo Savari crowd safety

**Solution** | CrowdGuard — real-time camera alert system

**Status** | Working prototype — ready to pilot